Email filtering and automation system

"how to take mails from email, outlook and differentiate between college and fraud mails and also link this system to a campus copilot" *Generated by LeapSpace*

Email Ingestion, Fraud Detection, and Campus Copilot Integration: Quick Insights

Key Findings Table

Topic	Key finding	Practical implication	Citations
Email ingestion	Institutional Outlook/Exchange access is moving from Basic Auth to Modern Authentication/OAuth 2.0, including IMAP/POP/SMTP scenarios.	Build ingestion around OAuth-based access rather than legacy password-based retrieval.	[1]
Secure mail operations	Centralized identity and filtering can reduce spam and improve traceability in institutional mail systems.	Put ingestion behind campus identity controls and gateway logging.	[2]
Fraud detection	Hybrid detectors that combine content, headers, URL/domain signals, and behavioral/rule signals consistently outperform single-signal approaches.	Use a layered rule-plus-ML pipeline, not a single classifier.	[3] [4] [5] [6] [7] [8]
Performance	Strong phishing studies report very high performance, including 99.8% F1 on all-email-component features and F1=0.9942 on an imbalanced phishing dataset.	High-quality fraud screening is feasible if training data and evaluation are realistic.	[6] [9]
College-mail routing	Research is strong on phishing/spam detection but weak on campus-specific routing into categories like admissions, registrar, finance, or housing.	Expect to build your own labeled institutional routing layer.	[10] [11] [12] [13] [14]

Topic	Key finding	Practical implication	Citations
Summarization/triage	Educational email classification and summarization can reduce handling burden, but human review remains important because automatic metrics do not always match human judgment.	Use summaries for copilot display, but keep staff review for low-confidence cases.	[15] [16] [17]
Privacy	S/MIME and related secure-email approaches improve confidentiality but can reduce server-side inspectability.	Design separate pathways for metadata screening and deeper content analysis where permitted.	[18] [19] [20] [21] [22] [23]
Copilot integration	Direct published evidence on campus-copilot integration for email triage is largely absent.	Treat copilot integration as an architectural extension built on top of proven detection and triage components.	[3] [15] [16] [17] [24]

Direct Answer

The best-supported way to do this is a **three-stage system**:

Ingest mail securely from Outlook/Exchange using **Modern Authentication/OAuth 2.0**, not Basic Auth; collect headers, sender/domain, subject, body text, URLs, attachment metadata, and available authentication signals. [1]

Classify fraud first using a **hybrid detector** that combines rules and machine learning across content, headers, URL/domain cues, and sender/network anomalies such as SPF/DKIM/DMARC issues, suspicious domains, foreign IPs, or protocol anomalies. Hybrid systems are the most consistently supported pattern in the evidence. [3] [4] [5] [6] [7] [8]

Route legitimate mail into campus categories such as admissions, registrar, finance, or IT support using a second classifier, then send the result to a **campus copilot** with a summary, risk score, label, and suggested action. The literature supports email classification and summarization, but not a ready-made campus-copilot implementation, so this part will likely need local design and labeling. [10] [11] [12] [15] [16] [17]

A practical architecture is therefore a **dual-head workflow**: **fraud-risk detection first**, **campus-service routing second only for low-risk mail**. Taken together, the evidence suggests this reduces the chance that phishing emails are accidentally treated as trusted college communications. [3] [4] [9]

Confidence level: High - the fraud-detection and secure-access parts are well supported; **Medium** for campus routing and copilot integration because direct studies are sparse. [1] [9] [10] [24]

Study Scope

Time and topic scope: The evidence covers modern institutional email access, privacy-preserving email handling, phishing/spam detection, educational email triage, and limited school-domain email classification from the retrieved 2018-2025 literature set. [1] [17] [25]

Methods covered: rule-based filtering, supervised ML, deep learning, hybrid rule-plus-ML systems, topic modeling, summarization, and privacy-preserving encrypted classification. [3] [4] [7] [22] [26] [27]

Operational context: campus/institutional email environments, including authentication migration, gateway filtering, and educational email handling. [1] [2] [28]

Assumptions & Limitations

The evidence **does not include a detailed published Microsoft Graph API campus pipeline** with explicit preprocessing steps such as header parsing, body cleaning, attachment handling, language detection, and deduplication before classification. [1] [29]

The evidence base for **campus-specific legitimate-mail routing** is thin; most retrieved work focuses on binary phishing/spam detection rather than admissions/registrar/finance classification. [10] [11] [12]

The retrieved studies **do not meaningfully describe direct integration with a campus copilot/chatbot**, including retrieval orchestration and escalation workflows. [3] [24]

Very high reported accuracy/F1 values should be interpreted carefully because production performance depends on dataset realism, imbalance, and human validation. [9] [17] [30] [31]

Suggested Further Research

Build and evaluate a **campus-specific labeled dataset** for legitimate categories such as admissions, registrar, finance, housing, and IT support. [10] [12]

Study **copilot integration patterns** for explanation, escalation, and retrieval over institutional knowledge sources. [15] [24]

Test **privacy-preserving classification** strategies for encrypted institutional email without sacrificing phishing detection quality. [21] [22] [23]

Evaluate with **false-positive cost, routing error, and human trust**, not only accuracy/F1. [9] [17] [30]

Compare **metadata-only screening versus full-content analysis** under real campus privacy policies. [22] [32]

Full Report Body

Introduction

Your goal has three distinct technical parts: **take emails from Outlook/email systems**, **separate legitimate college mail from fraud**, and **connect the output to a campus copilot**. The evidence strongly supports the first two parts as an institutional mail-security pipeline, but the third part is better treated as a system-design layer built on top of proven detection and triage components rather than as an already standardized pattern. [1] [3] [24]

A practical implementation should therefore be designed as a **secure ingestion layer**, a **fraud-detection layer**, and a **legitimate-mail routing/copilot layer**. This is consistent with the evidence showing mature practices for authentication modernization and hybrid phishing detection, alongside much weaker evidence for campus-service routing and copilot orchestration. [1] [9] [10] [24]

Confidence level: High - this architectural decomposition aligns closely with the retrieved evidence. [1] [3] [4]

Foundational Context

Before implementation, it helps to separate the problem into two classification tasks:

> **Fraud vs legitimate email**: identify phishing, impersonation, spam, and other malicious email using message content, header metadata, URLs, domain/authentication cues, and sender-recipient behavior. [3] [4] [6]
> **Legitimate college-mail routing**: once a message is judged low-risk, assign it to a campus function such as admissions, registrar, finance, or IT support. Direct literature here is limited, but educational email classification and broad domain labeling show the basic feasibility of such a second-stage classifier. [10] [12] [15] [16]

This distinction matters because the literature shows phishing detection is comparatively mature and high-performing, while campus business routing is not. A single model that tries to do both at once would likely mix two different objectives and increase operational risk. [9] [10] [30]

A second foundational issue is **privacy**. S/MIME and related secure-email designs protect message confidentiality and authenticity, but encrypted content limits what servers can inspect. This creates a direct tradeoff between confidentiality and automated analysis. [21] [22] [23]

Confidence level: High - both the two-stage classification framing and the privacy-inspection tradeoff are directly supported. [3] [10] [21]

Secure Email Ingestion and Privacy-Preserving Processing

Mail access pathways and authentication

Institutional email environments often use multiple protocols, including POP, IMAP, HTTP, and secure variants, and campus traffic studies show substantial use of cloud-hosted email services. [28] For modern deployment, the strongest operational signal is that institutions are retiring **Basic Authentication** and moving to **Modern Authentication/OAuth 2.0** for Exchange Online and IMAP/POP/SMTP access. Kyushu University explicitly reported this migration and its impact on Outlook and Thunderbird users. [1]

So, for your system, the ingestion layer should be designed around:

- OAuth-based access to institutional mailboxes, not password scraping or legacy Basic Auth. [1]
- Central identity enforcement, auditability, and mail gateway integration. [2]
- Protocol-agnostic collection where needed, because campuses may still have mixed client and service environments. [28]

A Zimbra/Proxmox Mail Gateway/Active Directory deployment showed that centralized filtering plus identity management can provide strong spam blocking, low false positives, and improved traceability. That does not prove the exact same result for Outlook/Exchange, but it does support the institutional value of a controlled gateway plus directory-backed identity architecture. [2]

Preprocessing before classification

The retrieved evidence does not provide a detailed Outlook/Graph preprocessing paper, but earlier email-processing systems used **content, headers, subject lines, clustering, social-network analysis, and supervised/unsupervised learning** for prioritization and grouping. [26] [27] Based on the supported feature patterns in fraud-detection studies, the preprocessing stage should at minimum extract:

- sender and recipient metadata [3] [4]
- subject and body text [6] [26]
- URLs and domains [4] [33]
- header/authentication cues such as SPF/DKIM/DMARC presence or absence when available [4]
- attachment-related features, because the strongest all-email-component model included header/body/attachment fields [6]

The gap is that the retrieved set does not explicitly describe detailed operational steps such as deduplication, attachment unpacking, or language detection for campus Outlook pipelines. [29]

Confidential handling and encryption-aware workflows

Secure email deployments may use **S/MIME**, smartcard-extended Outlook security, TPM-backed trust, and automated key handling. These approaches improve confidentiality, integrity, authentication, and nonrepudiation. [18] [19] [20] [21] However, encrypted mail reduces inspectability at the server side. [21] [23]

The evidence also shows a possible workaround: privacy-preserving spam filtering over encrypted content using homomorphic and functional encryption with leakage-reduction measures. [22] At the same time, university email protection systems were reported as weak at filtering incoming mail and phishing links, underscoring that confidentiality alone is not enough. [32]

A sensible institutional design is therefore:

- **gateway-level metadata/rule screening** for all messages [4] [8]
- **deeper content analysis only where institutionally authorized** [21] [23]
- **copilot access to labels/summaries rather than raw content** for privacy-sensitive messages, which is an inference consistent with the privacy findings. [17] [22]

Confidence level: Medium-High - authentication modernization is directly supported; detailed campus preprocessing is less documented. [1] [29]

Fraud and Phishing Detection for Institutional Email

Feature engineering

The evidence is strongest here. High-performing fraud detectors combine multiple signal types rather than relying on text alone. Supported feature groups include:

- **email/header/body features** [6]
- **URL and domain patterns** [4] [33]
- **HTML content and structural features** [4]
- **authentication and infrastructure cues**, including missing SPF/DKIM/DMARC, insecure protocols, and foreign IPs [4]
- **behavioral/sender-recipient signals** [3]
- **semantic or expert-derived text features**, including LLM-derived features in one study [34]

One study reported **99.8% F1** on IWSPA-AP using **114 engineered features plus TF-IDF over header/body/attachment fields**, which is the clearest feature-rich result in the evidence. [6]

Hybrid rule-based and machine-learning models

The evidence consistently favors **hybrid systems**. Examples include CNN+XGBoost with semantic-statistical features, heuristic-plus-ML multi-domain fusion, TF-IDF/heuristic ensembles, rule-preprocessing-plus-classifier systems, and voting/score-based architectures. [3] [4] [5] [7]

Rule layers matter because they improve coverage and explainability and can reduce false negatives and false positives. MailDefender explicitly adds a rule-based layer for this reason, and SpamAssassin-style systems remain a useful model: multiple weighted rules fire, the weights are summed, and a threshold determines spam-vs-ham classification. [8] [35]

For your system, this supports a workflow such as:

fast rules for obviously suspicious items

ML classifier for nuanced fraud detection

risk thresholding + human review for uncertain cases [4] [8] [35]

Performance patterns

Reported performance is strong:

MailDefender RF: **97.67% accuracy** [35]

soft-voting ensemble: **99.25% accuracy** and **0.9846 MCC** [5]

CNN-XGBoost: **0.9587 F1** [3]

THEMIS on unbalanced data: **99.848% accuracy** with **0.043% FPR** [30]

HELPHED on an imbalanced dataset: **F1=0.9942** [9]

Advanced 1D-CNNPD + Bi-GRU: **99.66% F1**, **99.68% accuracy**, **99.32% recall**, **100% precision** [25]

GCN+NLP: **98.2% accuracy** and **0.015 false-positive rate** [36]

The strongest methodological signal is not just high accuracy, but use of **imbalanced datasets** and **richer feature fusion**, which better reflects real mail streams. [9] [30] [37]

Confidence level: High - this is the best-supported part of the evidence base. [3] [6] [9]

College Email Classification and Triage

What the literature does and does not support

Here the evidence becomes much thinner. No retrieved study explicitly classifies college emails into the exact service categories you likely need, such as admissions, registrar, finance, financial aid, housing, IT helpdesk, and student support. [10] [12] [13] [14]

Closest matches include:

- a school-email system classifying only **General Inquiry vs Verification** using rule-based labeling plus GloVe-based classifiers, where KNN-GloVe performed best [10]
- a semantic spam study that first grouped email into broad domains such as **education, health, finance** before spam detection [12]
- an ontology-based classifier for **speech acts**, not campus service functions [11]

So the evidence does support the **feasibility** of a second-stage legitimate-mail classifier, but not an off-the-shelf campus taxonomy. [10] [12]

Educational prioritization and summarization

Educational email triage research is more encouraging. One study used deep learning with keyword extraction to classify student emails and reported **93% accuracy**. [16] A follow-up added email summarization and retained overall **93% accuracy** while improving per-label balance. [15] EMAILSUM introduced **2,549 human-annotated threads** and emphasized that human review is important because automatic metrics such as ROUGE and BERTScore were weakly correlated with human judgment. [17]

For your use case, that suggests:

- use summarization to present legitimate campus mail inside the copilot [15] [17]
- use human-reviewed labels to build the routing taxonomy [10] [17]
- expect iterative improvement rather than one-shot model training [15] [16]

Human review and labeling strategy

Because direct campus-category evidence is sparse, a human-in-the-loop labeling process is especially important. The school-email paper used rule-based labeling with keyword and dictionary support, showing that weak supervision can bootstrap a dataset. [10] Educational summarization work also reinforces the value of human evaluation over fully automatic scoring. [17]

A practical local strategy is:

- start with a campus category schema
- bootstrap labels with keywords/rules
- validate with staff reviewers
- train a second-stage classifier on the reviewed set

This last step is an inference from the evidence rather than a directly published campus standard, but it is the most evidence-aligned path. [10] [17]

Confidence level: Medium-Low - feasibility is supported, but direct campus routing evidence is limited. [10] [12]

Campus Copilot Integration and Operational Workflow

The retrieved papers do **not** provide a direct campus-copilot email integration blueprint. [3] [24] Still, the evidence supports the building blocks needed for one: classification, summarization, explainability, and human review. [15] [17] [31]

Copilot-facing routing design

A strong design, inferred from the evidence, is for the classifier pipeline to pass the following structured outputs to the campus copilot:

- fraud risk label and score [3] [35]
- legitimate campus-service label if low risk [10]
- short summary of the email or thread [15] [17]
- explanation features such as suspicious domain, missing SPF/DKIM/DMARC, or sender anomaly [4] [31]

The copilot should function mainly as an **explanation and retrieval layer**, not as the primary detector, because the evidence for core detection is much stronger than the evidence for conversational handling. [3] [24] [31]

Escalation and response handling

For suspected fraud, the safest evidence-aligned workflow is:

- quarantine or isolate high-risk messages [8] [35]
- warn the user with a concise explanation [31]
- escalate uncertain cases to IT/security staff or departmental reviewers [17] [35]

Detailed published escalation policies were not found, so this is a design inference anchored in hybrid scoring and human-review findings. [38] [39] [40]

Trust and usability

A 2024 web-based AI platform combined strong phishing performance with explainable AI, supporting the importance of transparent outputs in deployed systems. [31] The CoPilot secure email system also showed that usability matters even when cryptographic functions are automated. [41]

So the campus copilot should expose:

- why an email was flagged [31]
- what action is recommended [35]
- which campus department likely owns a legitimate request [10]

when human review is needed [17]

Confidence level: Medium-Low - this section is a careful system synthesis because direct copilot studies were not retrieved. [17] [24]

Evaluation, Deployment, and Implementation Considerations

Evaluation metrics

The evidence strongly supports evaluating fraud detection with:

accuracy [25] [30]

precision [25]

recall [25]

F1 [9] [25]

MCC where available [5]

false-positive rate [30] [36]

For your system, that should be expanded to include:

fraud false positives, because legitimate college mail wrongly blocked is operationally costly [30]

fraud false negatives, because missed phishing has security cost [35]

department routing accuracy for legitimate mail, once you build the second-stage classifier [10]

human acceptance/usability for summaries and copilot explanations [17] [31]

Datasets, imbalance, and generalization

The better phishing studies used realistic imbalance. THEMIS explicitly evaluated under unbalanced conditions. [30] HELPHED used **32,051 benign** and **3,460 phishing** emails and found soft voting superior to several baselines. [9] Another 2024 dataset paper released **seven curated phishing-email datasets totaling 203,176 instances**, reinforcing the value of broader feature-significance analysis for robust model building. [37]

For campus routing, however, public data appears sparse. One university-focused corpus collected **5,155 English-language scam emails from five large U.S. universities**, which may help for fraud-side adaptation but does not solve legitimate departmental routing. [29]

Institutional deployment constraints

Deployment should account for:

identity integration and access modernization [1]

gateway filtering and traceability [2]

privacy constraints in encrypted email flows [21] [23]

limited direct research on end-to-end Outlook ingestion architectures [29]

need for human review in summarization and ambiguous routing [17]

Confidence level: High for fraud evaluation principles; **Medium** for full campus deployment design because institutional routing and copilot integration remain under-studied. [9] [24]

Recommended Evidence-Based System Blueprint

A concise implementation blueprint, synthesized from the evidence, is:

Secure ingestion

Connect to Outlook/Exchange through Modern Authentication/OAuth 2.0. [1]
Route through centralized identity and mail gateway controls. [2]

Feature extraction

Parse headers, sender/recipient fields, subject/body text, URLs/domains, attachment metadata, and available mail-authentication indicators. [4] [6] [26]

Fraud screening

Apply a rule layer for obvious malicious indicators and threshold scoring. [8]
Apply an ML/hybrid classifier using combined content, metadata, semantic, and infrastructure features. [3] [4] [5] [34]

Legitimate-mail routing

Only for low-risk emails, run a second classifier that predicts campus function. This will likely require your own taxonomy and labeled institutional dataset. [10] [12]

Copilot integration

Provide summary, risk score, category, explanation, and recommended next action. [15] [17] [31]

Human-in-the-loop

Send low-confidence, high-risk, or privacy-sensitive cases to staff review. [17] [35]

Taken together, these findings suggest the most reliable near-term solution is not a single universal model, but a **layered institutional workflow** with clear separation between security detection, business routing, and user-facing copilot assistance. [2] [3] [9]

Conclusion

Yes, you can build this system, and the evidence points to a clear best path: **secure Outlook ingestion via Modern Authentication**, **hybrid fraud detection using multiple signals**, **a separate campus-mail routing model for legitimate mail**, and **a campus copilot that explains, summarizes, and escalates rather than replacing the classifier**. [1] [3] [4] [15] [17]

What is well proven is the fraud-detection core. What is not yet well covered in the literature is the campus-specific routing and copilot linkage, so those parts should be treated as institution-specific engineering and labeling work rather than as solved research problems. [9] [10] [24]

Overall confidence: Medium-High - strong support for secure ingestion and phishing detection; moderate support for educational triage; limited direct evidence for campus-category routing and copilot integration. [1] [9] [17] [24]

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